

## Cor 3 Puntos Recta Hiperbolica

**Corolario 1.** *Tres puntos  $z_1, z_2, z_3 \in \mathbb{D}$  distintos están en la misma recta hiperbólica si y sólo si  $\tau_{z_1}(z_3)/\tau_{z_1}(z_2) \in \mathbb{R}$ .*

***Demostración:*** Por definición,  $z_1, z_2, z_3$  están en la misma recta hiperbólica si y sólo si existe  $t \in \mathbb{R}$  tal que  $z_3 = \tau_{z_1}(\tau_{z_1}(z_2) t)$ . Aplicando  $\tau_{z_1}$  a ambos lados, por el **lem-involucion-disco-unidad**, esto es equivalente a que  $\tau_{z_1}(z_3) = \tau_{z_1}(z_2) t$ . Como  $z_2 \neq z_1$  y, por tanto,  $\tau_{z_1}(z_2) \neq 0$ , esto equivale a que  $\frac{\tau_{z_1}(z_3)}{\tau_{z_1}(z_2)} = t \in \mathbb{R}$ . ■

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